

Question

The thermal decomposition of a basic carbonate of a certain main group element is as follows: Accurately weigh 1.000 g of the salt, heat to about 300°C to undergo the first step of decomposition with a weight loss, producing two gases with a total weight of 0.1712g, cool to room temperature, and the remaining gaseous product is only 0.0941g. Continue to heat the remaining solid product, and the second step of decomposition with a weight loss occurs at about 425°C, and the mass of the remaining solid product is 0.5721g. Continue heating to constant weight, and the mass of the final product is measured to be 0.4310g. There are the following statements:

1. The element is +1 valence
2. The mass fraction of the metal in the basic carbonate is 0.2599
3. The final solid product remaining after heating is easily soluble in water
4. Obtain all the gases released during the above heating process, cool them to room temperature, and the mass of the remaining gas is 0.4919g
5. The number of gas molecules released in the second step of weight loss is 3 times the number of gas molecules released upon heating to constant weight; the number of gas molecules released in the first step of weight loss is 2 times the number of gas molecules released upon heating to constant weight.
6. The solid product of the second step of weight loss contains 3 elements

Select the option below that contains the most correct options

A. 1,2,3

B. 2,3,5

C. 1,3,5

D. 2,5,6

E. 1,5,6

F. 3,4,6

G. 4,5,6

H. 2,4,6

I. 1,4,5

J. 1,2,4

K. 2,3,4

Answer

Correct Answer: **D**

Detailed Explanation

Cooled to room temperature, the mass of the gaseous product decreased, obviously due to H_2O . Then the generated water vapor $m(\text{H}_2\text{O}) = 0.1712 - 0.0941 = 0.0771\text{g}$. The remaining gas from the thermal decomposition of the basic carbonate is CO_2 , then $n(\text{CO}_2) : n(\text{H}_2\text{O}) = 0.0941/44.01 : 0.0771/18.016 = 0.002138 : 0.004280 = 1 : 2$

CHECKPOINT

1 PTS

The portion of the gaseous product mass reduction upon cooling to room temperature is H_2O

CHECKPOINT

1 PTS

Calculated, the ratio of the number of molecules released of CO_2 to H_2O is 1:2

The second step weight loss $(1.000 - 0.5721) - 0.1712 = 0.2567\text{g}$, based on the amount of substance of CO_2 released in the first step weight loss, the corresponding relative molecular mass of the gas is $0.2567/0.002138 = 120.02$, corresponding to 3 H_2O and 1.5 CO_2

CHECKPOINT

1 PTS

Calculated, the second step weight loss produces 3 H₂O and 1.5 CO₂

Similarly, the third step weight loss $0.5721 - 0.4310 = 0.1411\text{g}$, corresponding to 1.5 CO₂, then the molecular weight of the gas produced in the first step weight loss is twice that produced by heating to constant weight; then the molecular weight of the gas produced in the second step weight loss is 3 times that produced by heating to constant weight, statement 5 is correct.

CHECKPOINT

1 PTS

Calculated, the third step weight loss produces 1.5 CO₂

The final product should be an oxide. Let the molecular weight of the oxide be x . A total of 4 CO₂ and 5 H₂O are lost. According to the question, the equation is $nx/(nx + 4 \times 44.01 + 5 \times 18.016) = 0.4310/1.000$. Simplifying, we get $x = 201.6/n$. When $n = 5$, there is a reasonable solution, corresponding to the oxide MgO. MgO is not easily soluble in water, statement 3 is incorrect.

CHECKPOINT

1 PTS

Calculated, the oxide is MgO, MgO is not easily soluble in water

Therefore, the chemical formula of the substance is 5MgO · 4CO₂ · 5H₂O, the magnesium element content is 0.2599 and it is +2 valent, statement 2 is correct, statement 1 is incorrect.

CHECKPOINT

1 PTS

Inferred that the chemical formula of the basic carbonate is $5\text{MgO} \cdot 4\text{CO}_2 \cdot 5\text{H}_2\text{O}$, containing metal 0.2599 and it is +2 valent

A total of 2.5 CO_2 and 5 H_2O were released after the second step weight loss. The remaining solid composition is $5\text{MgO} \cdot 1.5\text{CO}_2$, containing three elements, statement 6 is correct.

CHECKPOINT

1 PTS

After the second step weight loss, all H_2O is released, and the remaining solid composition is $5\text{MgO} \cdot 1.5\text{CO}_2$, containing three elements

After the released gas returns to room temperature, only CO_2 can remain gaseous. A total of $1.000/[5 \times (24.31 + 16) + 4 \times 44.01 + 5 \times 18.016] \times 4 \times 44.01 = 0.3764\text{g}$ can be released, statement 4 is incorrect.

CHECKPOINT

1 PTS

The released CO_2 remains gaseous after returning to room temperature, with a total mass of 0.3764g

Select option D